

CSL210

OBJECT ORIENTED PROGRAMING



— LAB PROJECT:

Submitted By

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■ Introduction

The **Online Complaint Management System (CMS)** is a console-based application developed using **C++**. The system is designed to facilitate users in registering complaints and administrators in managing and resolving those complaints efficiently.

The system uses **file handling** for data persistence, making it simple and lightweight.

GitHub Repository

Source code: <https://github.com/Amaan-Akbar/Complaint-Management-System>

Introduction

The Online Complaint Management System (CMS) is a console-based C++ application that allows users to file, track, edit and delete complaints, while giving administrators full control to review, confirm, reject and analyse all submissions. Every action is persisted to plain-text files (users.txt, complaints.txt, audit.txt) immediately, so no data is ever lost on exit. The terminal interface uses ANSI colour codes (cyan borders, yellow prompts, green success, red errors, magenta admin highlights) for a polished, role-aware experience on both Windows and Linux/macOS.

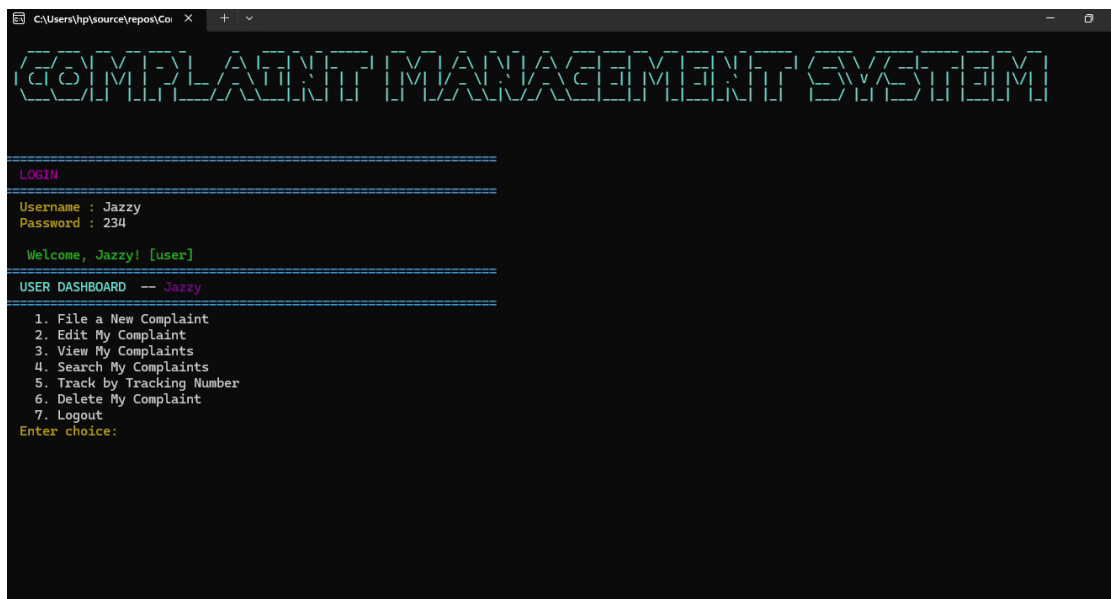


Figure 1 – ASCII banner, login screen, and User Dashboard

1. Objectives

- Develop a complete complaint handling system in C++ using OOP principles.
- Implement file-based persistence using `std::fstream` (users.txt / complaints.txt / audit.txt).
- Provide role-based access control: Admin and User, each with a dedicated dashboard.
- Issue unique CMS-XXXX tracking numbers so any complaint can be looked up at any time.
- Maintain an append-only audit trail that records actor, timestamp, and action for every change.
- Provide statistical reporting with proportional ASCII bar charts (status / priority / category).
- Demonstrate OOP concepts: class inheritance (Dashboard $\leftarrow$ Login), encapsulation, structs.
- Support cross-platform compilation (Windows: `localtime_s` + `SetConsoleMode`; Linux/macOS: `localtime_r`).

2. System Overview

The system operates with two main roles. Each role sees a completely separate menu tree.

User Features (7-option menu)

- File a New Complaint — choose category (7 options), priority (Low/Medium/High), enter description.
- Edit My Complaint — only editable while status is Pending and logged-in user is the owner.
- View My Complaints — filter by All / Pending / Confirmed / Rejected.
- Search My Complaints — case-insensitive keyword search across description and category.
- Track by Tracking Number — enter CMS-XXXX to retrieve full details of any owned complaint.
- Delete My Complaint — hard-delete a Pending complaint after y/n confirmation.
- Logout — returns to the login screen; system offers 'Login again? (y/n)'.

Admin Features (9-option menu)

- View / Filter All Complaints — filter by status or sort by priority (High first).
- Search Complaints — by keyword, category, or username.
- Confirm a Complaint — Pending → Confirmed, optional admin note.
- Reject a Complaint — Pending → Rejected, required rejection reason.
- Delete a Complaint — soft-delete (status = Deleted), hidden from normal views.
- Statistics Report — ASCII bar charts for status, priority, and category breakdowns.
- Audit Log — last 20 entries showing timestamp, actor (cyan/magenta), and action.
- Manage User Accounts — list, add (user or admin role), or delete non-admin users.
- Logout — returns to login screen.

3. Technologies Used

Component	Detail
Language	C++ (C++11 / C++17)
Paradigm	Object-Oriented Programming
Key OOP Concepts	Class inheritance, encapsulation, struct-based data modelling
File I/O	std::fstream — ifstream for read, ofstream for write/append
STL Containers	std::vector for users, complaints, and audit-log lists
String Processing	std::stringstream (pipe-delimited parsing), std::transform (case-insensitive search)
Formatting	std::ostringstream + setw + setfill for CMS-XXXX tracking numbers
UI	ANSI escape codes — colours, bold; Windows: SetConsoleMode(ENABLE_VIRTUAL_TERMINAL_PROCESSING)
Time	time_t / strftime — localtime_s (Windows) / localtime_r (POSIX)
Build	Any C++11-compatible compiler: g++, MSVC, clang++
Storage Files	users.txt · complaints.txt · audit.txt (plain text, human-readable)

4. System Design

4.1 Struct: AuditLog

Lightweight append-only record written to audit.txt after every key action.

Field / Method	Description
timestamp	Date-time string (YYYY-MM-DD HH:MM) from getCurrentTime().
actor	Username of the person who triggered the action.
action	Human-readable description, e.g. 'Filed complaint CMS-0007 [Traffic Violation]'.
save(ofstream&)	Appends one pipe-delimited line: timestamp actor action.
load(ifstream&) [static]	Reads and parses one pipe-delimited line from audit.txt.

4.2 Struct: Complaint

Stores all nine complaint fields and provides display helpers and I/O methods.

Field / Method	Description
id	Auto-incremented integer key; set to max existing id + 1 on creation.
trackingNo	CMS-XXXX (zero-padded to 4 digits). Generated by makeTrackingNo(id).
status	State machine value: Pending Confirmed Rejected Deleted.
description	Free-text entered by the user via getline().
category	One of 7 fixed options selected from a numbered menu (pickCategory()).
priority	Low Medium High — from pickPriority() numbered menu.
username	Owner's login name, set automatically from loggedUser at filing time.
timestamp	Filing date-time from getCurrentTime() — not updated on edit.
adminNote	Set by admin on Confirm ('Reviewed and confirmed.') or Reject (required reason).
save(ofstream&)	Writes all 9 fields as consecutive lines to complaints.txt.
load(ifstream&) [static]	Reads 9 consecutive lines; returns false if id line missing.
statusBadge()	Returns ANSI-coloured badge string: yellow PENDING, green CONFIRMED, red REJECTED/DELETED.
priorityLabel()	Returns ANSI-coloured label: red HIGH, yellow MEDIUM, green LOW.
printSummary()	Prints a full formatted complaint card (tracking no, badge, priority, category, owner, timestamp, desc, admin note).

4.3 Struct: User

Field / Method	Description
username	Unique login name; duplicates rejected by addUser().
password	Plain-text password stored in users.txt.
role	'admin' or 'user' — determines which dashboard is shown after login.
save(ofstream&)	Writes one pipe-delimited line: username password role.
load(ifstream&) [static]	Reads and parses one line; returns false if empty.

4.4 Class: Login

Owns vector<User> users. Manages authentication, first-run setup, and user account CRUD. Protected members (loggedRole, loggedUser) are inherited by Dashboard.

- loadUsers() / saveUsers() — read/write users.txt on startup and every mutation.
- hasUsers() — returns false on first run; triggers registerAccounts().
- registerAccounts() — interactive wizard creating one admin + one user; saves immediately.
- login() — validates username+password; sets loggedRole and loggedUser on success.
- logout() — clears session strings.
- addUser() — checks for duplicate username before inserting; saves to disk.
- deleteUser() — blocks deletion of admin-role accounts; saves to disk.
- listUsers() — prints all accounts with role colour coding (magenta=admin, cyan=user).

4.5 Class: Dashboard (inherits Login)

The main application controller. Inherits all authentication behaviour and adds complaint management, statistics, audit logging, and the runSystem() entry point.

- loadComplaints() / saveComplaints() — read/write complaints.txt; tracks max id.
- log(actor, action) — creates AuditLog entry, appends to audit.txt (ios::app).
- pickMenu(min,max) — input loop with cin.fail() guard and clearInput() recovery.

- pickCategory() / pickPriority() — numbered sub-menus returning string values.
- makeTrackingNo(id) — formats CMS-XXXX using ostream + setw(4) + setfill('0').
- findById(id) — linear scan of complaints vector, returns pointer or nullptr.
- registerComplaint() — full filing wizard; auto-assigns id, trackingNo, status, username, timestamp.
- editComplaint() — lists own Pending complaints; allows description-only or full category+priority+description edit.
- viewMyComplaints() — status filter (All/Pending/Confirmed/Rejected) on own complaints.
- searchComplaints() — case-insensitive substring search across description and category.
- trackComplaint() — looks up any complaint by CMS-XXXX number.
- deleteMyComplaint() — hard-erases from vector (erase+remove_if) after y/n confirm; Pending only.
- adminViewComplaints() — all complaints (excluding Deleted) with filter + High-first sort.
- adminSearch() — search by keyword, category (uses pickCategory()), or username.
- confirmComplaint() / rejectComplaint() — status transitions from Pending only; save + log.
- adminDeleteComplaint() — soft-delete (sets status = 'Deleted'); saves + logs.
- showStatistics() — dynamic category counting + proportional ASCII bar chart rendering.
- viewAuditLog() — reloads audit.txt, shows last 20 entries.
- manageUsers() — sub-menu wrapping Login's addUser / deleteUser / listUsers.
- userDashboard() / adminDashboard() — role-specific infinite menu loops.
- runSystem() — entry point: Windows ANSI setup, load data, 3-attempt login loop.

5. Working of System

5.1 Startup & Login

On launch the ASCII banner is printed, then users.txt and complaints.txt are loaded into memory. If users.txt is absent the first-run wizard runs. After that the login loop begins with up to 3 attempts; failure exits the program. On success the user's role determines which dashboard is displayed.



```

C:\Users\hp\source\repos\Com... x + v
COMPLAINT MANAGEMENT SYSTEM

=====
LOGIN
=====
Username : Jazzy
Password : 234

Welcome, Jazzy! [user]

=====
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice:
  
```

Figure 2 – Startup banner → Login → Welcome message → User Dashboard

5.2 Filing a Complaint

User selects option 1 from the User Dashboard. The system presents a numbered category menu (7 options)

followed by a priority menu (Low / Medium / High), then prompts for a free-text description. The system auto-assigns the next id, formats the CMS-XXXX tracking number, sets status to Pending, records the owner and timestamp, saves to complaints.txt, and appends an audit entry.

```
C:\Users\hp\source\repos\Com X + v
Password : 234
Welcome, Jazzy! [user]
=====
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 1
=====
FILE A COMPLAINT
=====
Select category:
1. Theft / Robbery
2. Harassment
3. Fraud / Scam
4. Vandalism
5. Noise / Disturbance
6. Traffic Violation
7. Other
Enter choice: 6
Select priority:
1. Low
2. Medium
3. High
Enter choice: 2
Description: Viola
```

Figure 3 – Filing a complaint: category selection → priority → description

```
C:\Users\hp\source\repos\Com X + v
5. Noise / Disturbance
6. Traffic Violation
7. Other
Enter choice: 6
Select priority:
1. Low
2. Medium
3. High
Enter choice: 2
Description: Violation of signal
Complaint filed! Tracking No: CMS-0007
=====
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 2
=====
EDIT COMPLAINT
=====
Your pending complaints:
ID 2 | CMS-0002 | Harassment
ID 5 | CMS-0005 | cyber threat
ID 6 | CMS-0006 | Robbery in Bank
ID 7 | CMS-0007 | Violation of signal
Enter complaint ID to edit:
```

Figure 4 – Complaint filed confirmation showing CMS-0007 tracking number

5.3 Editing a Complaint

Option 2 lists all Pending complaints owned by the logged-in user. The user enters the numeric ID, then chooses to update description only, or category + priority + description. Only Pending complaints are shown; Confirmed or Rejected complaints cannot be edited.

```
C:\Users\hp\source\repos\Com X + v
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 2

=====
EDIT COMPLAINT
=====

Your pending complaints:
ID 2 | CMS-0002 | Harassment
ID 5 | CMS-0005 | cyber threat
ID 6 | CMS-0006 | Robbery in Bank
ID 7 | CMS-0007 | Violation of signal

Enter complaint ID to edit: 5
What do you want to change?
1. Description only
2. Category and Priority too
Enter choice: 2

Select category:
1. Theft / Robbery
2. Harassment
3. Fraud / Scam
4. Vandalism
5. Noise / Disturbance
6. Traffic Violation
7. Other
Enter choice: 3

Select priority:
1. Low
2. Medium
3. High
Enter choice: |
```

Figure 5 – Edit Complaint: listing pending items, selecting what to change

5.4 Viewing & Tracking Complaints

Option 3 (View My Complaints) lets the user filter by status. Each complaint is printed as a formatted card showing the tracking number, colour-coded status badge (YELLOW=Pending, GREEN=Confirmed, RED=Rejected), priority label, category, owner, timestamp, description, and any admin note.

```
C:\Users\hp\source\repos\Com X + v

#CMS-0003 [CONFIRMED] Priority: LOW Category: Traffic Violation
Filed by: Jazzy at 2026-05-20 01:24
Desc : Signal Violation
Admin note: Reviewed and confirmed.

#CMS-0004 [REJECTED] Priority: LOW Category: Other
Filed by: Jazzy at 2026-05-20 01:25
Desc : Fraud
Admin note: Rejected by admin.

#CMS-0005 [PENDING] Priority: LOW Category: Fraud / Scam
Filed by: Jazzy at 2026-05-20 01:29
Desc : Online Fraud

#CMS-0006 [PENDING] Priority: MEDIUM Category: Theft / Robbery
Filed by: Jazzy at 2026-05-20 01:40
Desc : Robbery in Bank

#CMS-0007 [PENDING] Priority: MEDIUM Category: Traffic Violation
Filed by: Jazzy at 2026-05-20 01:42
Desc : Violation of signal

Total: 7 complaint(s).

=====
USER DASHBOARD -- Jazzy
=====

1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 4
```

Figure 6 – Complaint list showing CONFIRMED, REJECTED, and PENDING badges with admin notes

Option 4 (Search) performs a case-insensitive keyword match across description and category fields.


```
C:\Users\hp\source\repos\Com X + -
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 4

=====
SEARCH MY COMPLAINTS
Enter keyword: Violation

=====
#CMS-0003 [CONFIRMED] Priority: LOW Category: Traffic Violation
Filed by: Jazzy at 2026-05-20 01:24
Desc : Signal Violation
Admin note: Reviewed and confirmed.

#CMS-0007 [PENDING] Priority: MEDIUM Category: Traffic Violation
Filed by: Jazzy at 2026-05-20 01:42
Desc : Violation of signal

Total: 2 complaint(s).

=====
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 5

=====
TRACK COMPLAINT
Enter tracking number (e.g. CMS-0001): CMS-0
```

Figure 7 – Search results for keyword "Violation" — shows matching complaints

Option 5 (Track) lets the user enter a CMS-XXXX number and immediately see full complaint details.

```
C:\Users\hp\source\repos\Com X + -
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 5

=====
TRACK COMPLAINT
Enter tracking number (e.g. CMS-0001): CMS-0003

=====
#CMS-0003 [CONFIRMED] Priority: LOW Category: Traffic Violation
Filed by: Jazzy at 2026-05-20 01:24
Desc : Signal Violation
Admin note: Reviewed and confirmed.

=====
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: 6

=====
DELETE MY COMPLAINT
Enter complaint ID to delete: |
```

Figure 8 – Track Complaint by CMS-0003 — shows Confirmed status with admin note

5.5 Deleting a Complaint (User)

Option 6 asks for the complaint ID, verifies ownership and Pending status, then prompts 'Are you sure? (y/n)'. On confirmation the entry is hard-erased from the vector, complaints.txt is rewritten, and an audit entry is appended.

```
C:\Users\hp\source\repos\Com X + v
Enter choice: 6
=====
DELETE MY COMPLAINT
=====
Enter complaint ID to delete: CMS-0002
Not found or not yours.
=====
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: Invalid! Enter a number 1-7.
Enter choice: 6
=====
DELETE MY COMPLAINT
=====
Enter complaint ID to delete: 2
Are you sure? (y/n): y
Complaint deleted.
=====
USER DASHBOARD -- Jazzy
=====
1. File a New Complaint
2. Edit My Complaint
3. View My Complaints
4. Search My Complaints
5. Track by Tracking Number
6. Delete My Complaint
7. Logout
Enter choice: |
```

Figure 9 – Delete flow: ownership guard ("Not found or not yours"), then successful delete with y/n confirm

5.6 Admin Dashboard

```
C:\Users\hp\source\repos\Com X + v
Filed by: Jazzy at 2026-05-20 01:40
Desc : Robbery in Bank
=====
#CMS-0007 [PENDING] Priority: MEDIUM Category: Traffic Violation
Filed by: Jazzy at 2026-05-20 01:42
Desc : Violation of signal
=====
#CMS-0003 [CONFIRMED] Priority: LOW Category: Traffic Violation
Filed by: Jazzy at 2026-05-20 01:24
Desc : Signal Violation
Admin note: Reviewed and confirmed.
=====
#CMS-0004 [REJECTED] Priority: LOW Category: Other
Filed by: Jazzy at 2026-05-20 01:25
Desc : Fraud
Admin note: Rejected by admin.
=====
#CMS-0005 [PENDING] Priority: LOW Category: Fraud / Scam
Filed by: Jazzy at 2026-05-20 01:29
Desc : OnLine Fraud
=====
Total: 5 complaint(s).
=====
ADMIN DASHBOARD -- Amaan
=====
1. View / Filter All Complaints
2. Search Complaints
3. Confirm a Complaint
4. Reject a Complaint
5. Delete a Complaint
6. Statistics Report
7. Audit Log
8. Manage User Accounts
9. Logout
Enter choice: |
```

Figure 10 – Admin logs in as "Amaan" — admin dashboard shows all 9 options

5.7 Confirm & Reject Complaints

Option 3 (Confirm) and Option 4 (Reject) both require the numeric complaint ID. Only Pending complaints can be transitioned. On Confirm the admin may add an optional note (defaults to 'Reviewed and confirmed.'). On Reject a reason is required (defaults to 'Rejected by admin.' if blank). Status is updated, complaints.txt saved, audit logged.

```
C:\Users\hp\source\repos\Com... X + -
ADMIN DASHBOARD -- Amaan
=====
1. View / Filter All Complaints
2. Search Complaints
3. Confirm a Complaint
4. Reject a Complaint
5. Delete a Complaint
6. Statistics Report
7. Audit Log
8. Manage User Accounts
9. Logout
Enter choice: 3
=====
CONFIRM COMPLAINT
=====
Enter complaint ID to confirm: 6
Add admin note (press Enter to skip):
Complaint CMS-0006 confirmed.
=====
ADMIN DASHBOARD -- Amaan
=====
1. View / Filter All Complaints
2. Search Complaints
3. Confirm a Complaint
4. Reject a Complaint
5. Delete a Complaint
6. Statistics Report
7. Audit Log
8. Manage User Accounts
9. Logout
Enter choice: 4
=====
REJECT COMPLAINT
=====
Enter complaint ID to reject:
```

Figure 11 – Confirming CMS-0006, then initiating a Reject on the next complaint

5.8 Statistics Report

Option 6 dynamically counts all complaints and renders proportional ASCII bar charts (# = filled, - = empty, width=20) across three dimensions. Categories are discovered at runtime — no hardcoded list. Bar width is proportional to total complaints.

```
C:\Users\hp\source\repos\Com... X + -
STATISTICS REPORT
=====
STATUS BREAKDOWN
[###-----] Pending : 1
[#####-----] Confirmed : 2
[#####-----] Rejected : 2
[###-----] Deleted : 1
Total : 6

PRIORITY BREAKDOWN
[###-----] High : 1
[#####-----] Medium : 2
[#####-----] Low : 3

BY CATEGORY
[#####-----] Theft / Robbery : 2
[#####-----] Traffic Violation : 2
[###-----] Other : 1
[###-----] Fraud / Scam : 1
=====
ADMIN DASHBOARD -- Amaan
=====
1. View / Filter All Complaints
2. Search Complaints
3. Confirm a Complaint
4. Reject a Complaint
5. Delete a Complaint
6. Statistics Report
7. Audit Log
8. Manage User Accounts
9. Logout
Enter choice:
```

Figure 12 – Statistics Report: status, priority, and category breakdowns with ASCII bars

5.9 Audit Log

Option 7 reloads audit.txt and displays the last 20 entries. Each line shows the timestamp in cyan, the actor's username in magenta, and the action in white. The log below records a complete demo session.

```

C:\Users\hp\source\repos\Co... X + -
2026-05-20 01:26 Jazzy -> Edited complaint CMS-0001
2026-05-20 01:30 Jazzy -> Filed complaint CMS-0005 [Other]
2026-05-20 01:31 Amaan -> Confirmed complaint CMS-0003
2026-05-20 01:32 Amaan -> Rejected complaint CMS-0004 -- Rejected by admin.
2026-05-20 01:33 Amaan -> Admin deleted complaint CMS-0001
2026-05-20 01:34 Amaan -> Created account: Iftikhar [admin]
2026-05-20 01:40 Jazzy -> Filed complaint CMS-0006 [Theft / Robbery]
2026-05-20 01:42 Jazzy -> Filed complaint CMS-0007 [Traffic Violation]
2026-05-20 01:43 Jazzy -> Edited complaint CMS-0005
2026-05-20 01:45 Jazzy -> Deleted complaint CMS-0002
2026-05-20 01:46 Amaan -> Confirmed complaint CMS-0006
2026-05-20 01:47 Amaan -> Rejected complaint CMS-0007 -- Not verified

=====
ADMIN DASHBOARD -- Amaan
=====
1. View / Filter All Complaints
2. Search Complaints
3. Confirm a Complaint
4. Reject a Complaint
5. Delete a Complaint
6. Statistics Report
7. Audit Log
8. Manage User Accounts
9. Logout
Enter choice: 8

=====
MANAGE USERS
=====
1. List all users
2. Add new user
3. Delete a user
4. Back
Enter choice: 2
New username : Anis

```

Figure 13 – Audit Log showing 16 entries: logins, filings, edits, confirms, rejects, deletes

5.10 Manage User Accounts

Option 8 opens a sub-menu with three operations. List shows all accounts with role colour coding. Add prompts for username, password, and role (1=user / 2=admin); duplicate names are rejected. Delete checks the user is not admin and not the currently logged-in admin (self-delete guard).

```

C:\Users\hp\source\repos\Co... X + -
9. Logout
Enter choice: 8

=====
MANAGE USERS
=====
1. List all users
2. Add new user
3. Delete a user
4. Back
Enter choice: 2
New username : Anis
Password :
Role (1=user, 2=admin): 1
User added.

=====
MANAGE USERS
=====
1. List all users
2. Add new user
3. Delete a user
4. Back
Enter choice: 4

=====
ADMIN DASHBOARD -- Amaan
=====
1. View / Filter All Complaints
2. Search Complaints
3. Confirm a Complaint
4. Reject a Complaint
5. Delete a Complaint
6. Statistics Report
7. Audit Log
8. Manage User Accounts
9. Logout
Enter choice: |

```

Figure 14 – Manage Users: adding a new user account "Anis" with role=user

6. Complaint Lifecycle

State	Who Sets It	Condition	Notes
Pending	System (auto)	On creation	Initial state for every new complaint.
Confirmed	Admin	Must be Pending	Optional admin note; default 'Reviewed and confirmed.'
Rejected	Admin	Must be Pending	Rejection reason required; stored in adminNote.
Deleted	Admin	Any status	Soft-delete; excluded from normal views but kept

(Removed)	User (hard)	Must be Pending + owner	in file. erase(remove_if) — permanently removed from complaints.txt.
-----------	-------------	-------------------------	---

7. Input Validation

Mechanism	Implementation Detail
Menu range check	pickMenu(min,max) loops indefinitely until a valid int in [min,max] is entered.
Stream error recovery	cin.fail() → cin.clear() + cin.ignore(max,\n) flushes bad input.
Mixed input guard	clearInput() called before every getline() to consume the trailing \n left by cin >>.
3-attempt login lockout	Counter in runSystem() loop; 4th failure prints error and breaks out.
Ownership guard	Edit/delete verify c->username == loggedUser before proceeding.
Status guard	Edit/user-delete: only Pending. Confirm/reject: only Pending.
Self-delete guard	manageUsers() rejects delete if target username == loggedUser.
Admin-role guard	deleteUser() refuses if it->role == 'admin'.
Duplicate username guard	addUser() scans existing users and returns false on collision.
Complaint ID not found	findById() returns nullptr; callers print 'Not found.' and return.

8. Key Features

Feature	Description
ASCII Banner	Multi-line art title 'COMPLAINT MANAGEMENT SYSTEM' printed in bold cyan at startup.
Role-Based Dashboards	Separate 7-item (user) and 9-item (admin) menus; no cross-access at application layer.
Unique Tracking Numbers	CMS-XXXX zero-padded IDs, never reused (max tracked on load).
7 Complaint Categories	Theft/Robbery, Harassment, Fraud/Scam, Vandalism, Noise/Disturbance, Traffic Violation, Other.
3-Level Priority System	High (red) / Medium (yellow) / Low (green) — set at filing, changeable on edit.
Colour-Coded Status Badges	[PENDING] yellow · [CONFIRMED] green · [REJECTED] red · [DELETED] red.
Keyword Search	Case-insensitive substring match across description + category; admin can also search by username.
Audit Trail	Append-only log: timestamp actor action for every mutation.
ASCII Statistics	Dynamic category counting + proportional bar chart rendering in showStatistics().
Immediate Persistence	Every mutation flushes to disk instantly; no deferred writes or data loss on exit.
First-Run Setup	Interactive wizard creates admin + user account when users.txt is absent.
Login Retry + Lockout	3 attempts; 4th failure exits the program with a clear error message.
Cross-Platform	Windows: SetConsoleOutputCP + SetConsoleMode. POSIX: localtime_r. Same source.

9. Limitations

- Passwords stored as plain text in users.txt — no hashing or salting.
- File-based storage has no concurrency control; simultaneous multi-user access would corrupt data.
- No pagination — large complaint lists are printed in full to the terminal.
- Complaint IDs are never recycled; gaps appear after user hard-deletes.

- No email or push notification when complaint status changes.
- Console-only interface — no GUI, web front-end, or mobile app.
- Single admin privilege level — no per-admin permission scoping.
- Audit log is append-only with no built-in rotation or size limit.

10. Future Enhancements

Enhancement	Description
Password Hashing	Integrate bcrypt or SHA-256 to secure stored credentials.
Database Backend	Replace flat files with SQLite or MySQL for concurrent multi-user access.
GUI / Web Front-end	Port the logic to Qt desktop app or REST API + React web UI.
Email Notifications	Alert users automatically when their complaint status changes.
Pagination	Display a fixed number of records per screen for large datasets.
Report Export	Export statistics to CSV or PDF for offline analysis and sharing.
Two-Factor Auth (2FA)	Add TOTP-based second factor for admin accounts.
Complaint Reassignment	Allow admins to transfer ownership between users.
Advanced Search & Filters	Combined filters: status + category + priority + date range.
Audit Log Rotation	Limit audit.txt size; archive old entries automatically.

11. Conclusion

The Complaint Management System successfully demonstrates Object-Oriented Programming and file handling in C++. The two-class hierarchy (Dashboard inherits Login) cleanly separates authentication concerns from application logic, while the three struct types (User, Complaint, AuditLog) keep each data entity self-contained with its own I/O methods.

As shown in the demo video, the system handles a complete real-world workflow: user Jazzy files multiple complaints across different categories and priorities; admin Amaan views, confirms, rejects, and deletes complaints; the audit log records every action with timestamps; and the statistics report visualises the data with colour-coded ASCII bar charts. All changes are flushed to disk immediately, making the system resilient to unexpected exits.

The system provides a solid, well-structured foundation that can be extended into a full-scale application by integrating a database back-end, a GUI or web front-end, proper password hashing, and email notifications.